Dohyun (Eric) Kim

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PROFILE

Penultimate Computer Systems Engineering (Hons) student at the University of Auckland, specialising in embedded systems and digital hardware design. AWS certified in both Cloud and AI/ML, with a competition-winning project portfolio spanning FPGA development, PCB design, and embedded robotics systems.

EDUCATION

University of Auckland - Auckland CBD, New Zealand *Expected Graduation Nov 2027*

- Degree: *Bachelor's (Hons) of Engineering in Computer Systems Engineering*
- Concentration: Embedded Systems & Digital Hardware Design

EXPERIENCE

Teaching Assistant **Auckland, New Zealand**
University of Auckland *Jun 2026 - Present*

- Delivered teaching support across tutorials, lab sessions, and test invigilation for **ELECTENG 292: Electronics** to strengthen **student leadership** and technical laboratory techniques.

Robotics Instructor **Auckland, New Zealand**
Creative Imaginary Lab (ciLab) *Apr 2026 - Present*

- Instructed students in robot **hardware assembly** and **software programming** to complete missions, while coaching and supporting teams in preparation for **nationwide robotics competitions**.

PROJECTS

Flappy Universe **VHDL**
FPGA Game Implementation - Team Project *Jun 2026*

- Implemented a Flappy Bird-style game in VHDL on an **Altera FPGA** with VGA output and sprite rendering in order to demonstrate real-time hardware design.
- Developed a pixel-priority VHDL pipeline in order to render game scenes at VGA resolution.

Smart Energy Monitor **C & Embedded**
Embedded Energy Monitoring System - Team Project *Oct 2025*

- Designed an embedded energy monitoring system using **ATmega328P** microcontrollers with ADC processing in order to measure and display real-time energy usage.
- Produced a validated PCB schematic using **Altium Designer** in order to achieve accurate signal conditioning.

AWARDS

3rd Place in 2025 ECSE Design Competition **Team Project**
"Winnie the Bot" - Interactive companion robot *Sep 2025*

- Built an AI-powered robot using dual **ATmega328P** microcontrollers, an AI camera, and audio peripherals in order to enable face tracking, arm movement, and voice dialogue in a compact form factor.
- Contributed to **prototyping** the enclosure using iterative design in order to deliver a functional physical build.

CERTIFICATIONS

AWS Certified Cloud Practitioner **Cloud Computing**
Amazon Web Services (AWS) *Apr 2026*

- Validated foundational knowledge of **AWS cloud concepts**, core services, and cloud security and architecture.

AWS Certified AI Practitioner **AI/ML**
Amazon Web Services (AWS) *May 2026*

- Validated foundational knowledge of AI/ML concepts, generative AI, and **AWS AI/ML services** and tools.

ACTIVITIES & LEADERSHIP

Academic Team Executive **Auckland, New Zealand**
KEB (Korean Engineering Body) *Jul 2024 - Present*

- Tutored junior engineering students and assisted in planning academic events for the student community.

Full time Volunteer **Auckland, New Zealand**
IEEE (Institute of Electrical and Electronics Engineers) *Jul 2025*

- Volunteered 40+ hours full-time supporting competition operations at the nationwide NZ Robotics Olympiad 2025.

SKILLS

Languages & Frameworks: C, VHDL, MATLAB, Python, Java, JavaScript, React.js, Pandas, Scikit-Learn
Tools: Altium Designer, LTspice, Proteus, Intel Quartus, Atmel AVR, AutoCAD, AWS, Git/GitHub